

# Moving Beyond Right Triangles to the Unit Circle

MATHEMATICS LECTURE NOTES • TRIGONOMETRY FOUNDATIONS

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## 1. The Right-Triangle Limitation

Most people are introduced to trigonometry via the classic mnemonic **SOH-CAH-TOA**. This geometric framework operates perfectly fine as long as you are constrained within the bounds of a right-angled triangle.

But what happens when the angle  $\theta$  expands to  $120^\circ$ ? Or drops to a negative value like  $-45^\circ$ ? It is physically impossible to construct a Euclidean right triangle with a negative angle, nor can an interior angle exceed  $90^\circ$  alongside a right angle. To resolve these real-world and theoretical limitations, we must scale up our paradigm using the **Unit Circle**.

## 2. The Unit Circle Framework

The unit circle is defined simply as a circle centered at the Cartesian origin  $(0,0)$  with a radius of exactly **1**. The algebraic equation for this locus of points is:

$$x^2 + y^2 = 1$$

If we place an angle  $\theta$  in standard position—where the initial side lies along the positive x-axis and rotation proceeds counterclockwise—the terminal side intersects the unit circle circumference at a unique coordinate point  $(x, y)$ .

### The Universal Definition

- $\cos \theta = x$
- $\sin \theta = y$
- $\tan \theta = \frac{y}{x}$  (where  $x \neq 0$ )

Because the radius  $r = 1$ , the traditional hypotenuse denominator drops completely out of the trigonometric ratios. Consequently, the coordinates themselves directly represent the cosine and sine values.

### 3. Deriving the Pythagorean Identity

Since any point  $(x, y)$  on the terminal side rests directly on the circumference of the circle, its coordinates must identically satisfy the circle's equation  $x^2 + y^2 = 1$ .

By substituting our universal coordinate-based definitions into this equation, we elegantly derive the fundamental identity of all trigonometry:

$$\cos^2\theta + \sin^2\theta = 1$$

From this single foundational touchstone, we can systematically derive two additional Pythagorean identities by dividing the entire equation by either  $\cos^2\theta$  or  $\sin^2\theta$ :

- Dividing by  $\cos^2\theta$  yields:  $1 + \tan^2\theta = \sec^2\theta$
- Dividing by  $\sin^2\theta$  yields:  $\cot^2\theta + 1 = \csc^2\theta$

### 4. Standard Reference Values

In calculus and advanced physics applications, mathematical convention strongly favors **radians** over degrees. Below is a rigorous summary of standard first-quadrant reference values:

| Degree | Radian          | $\cos \theta$ (x-coord) | $\sin \theta$ (y-coord) | $\tan \theta$ (y/x)  |
|--------|-----------------|-------------------------|-------------------------|----------------------|
| 0°     | 0               | 1                       | 0                       | 0                    |
| 30°    | $\frac{\pi}{6}$ | $\frac{\sqrt{3}}{2}$    | $\frac{1}{2}$           | $\frac{\sqrt{3}}{3}$ |
| 45°    | $\frac{\pi}{4}$ | $\frac{\sqrt{2}}{2}$    | $\frac{\sqrt{2}}{2}$    | 1                    |
| 60°    | $\frac{\pi}{3}$ | $\frac{1}{2}$           | $\frac{\sqrt{3}}{2}$    | $\sqrt{3}$           |
| 90°    | $\frac{\pi}{2}$ | 0                       | 1                       | Undefined            |

### 5. Sign Conventions by Quadrant (ASTC)

As the terminal ray swept across the coordinate system moves through the four distinct quadrants, the algebraic signs of  $x$  and  $y$  transition, changing the signs of the respective functions. The mnemonic **ASTC** (*All Students Take Calculus*) maps out where functions remain **positive**:

- **Quadrant I** (Top-Right): All functions are positive, as both coordinates  $x > 0$  and  $y > 0$ .

- **Quadrant II (Top-Left):** Sine (and cosecant) is positive; because  $x < 0$ , cosine and tangent evaluate to negative values.
- **Quadrant III (Bottom-Left):** Tangent (and cotangent) is positive; both coordinates are negative ( $x < 0, y < 0$ ), meaning their ratio cancels out to become positive.
- **Quadrant IV (Bottom-Right):** Cosine (and secant) is positive; because  $y < 0$ , sine and tangent remain strictly negative.